

Ex. 20. 50 is divided into two parts such that the sum of their reciprocals is $\frac{1}{12}$. Find the two parts.

Sol. Let the two parts be x and $(50 - x)$.

$$\begin{aligned}\text{Then, } \frac{1}{x} + \frac{1}{50-x} &= \frac{1}{12} \Rightarrow \frac{50-x+x}{x(50-x)} = \frac{1}{12} \Rightarrow x^2 - 50x + 600 = 0 \\ &\Rightarrow (x-30)(x-20) = 0 \Rightarrow x = 30 \text{ or } x = 20.\end{aligned}$$

So, the parts are 30 and 20.

Ex. 21. If three numbers are added in pairs, the sums equal 10, 19 and 21. Find the numbers.

Sol. Let the numbers be x , y and z . Then,

$$x + y = 10 \quad \dots(i)$$

$$y + z = 19 \quad \dots(ii)$$

$$x + z = 21 \quad \dots(iii)$$

Adding (i), (ii) and (iii), we get : $2(x + y + z) = 50$ or $(x + y + z) = 25$.

Thus, $x = (25 - 19) = 6$; $y = (25 - 21) = 4$; $z = (25 - 10) = 15$.

Hence, the required numbers are 6, 4 and 15.

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions : Mark (✓) against the correct answer:

1. By how much is $\frac{3}{4}$ th of 568 lesser than $\frac{7}{8}$ th of 1008?

(Bank P.O., 2008)

- (a) 444 (b) 448
(c) 452 (d) 456
(e) None of these

2. The difference between a number and its three-fifths is 50. What is the number?

- (a) 75 (b) 100
(c) 125 (d) None of these

3. If a number is added to two-fifths of itself, the value so obtained is 455. What is the number?

(Bank Recruitment, 2010)

- (a) 325 (b) 350
(c) 400 (d) 420
(e) None of these

4. If a number is multiplied by two-thirds of itself the value so obtained is 864. What is the number?

(Bank Recruitment, 2010)

- (a) 34 (b) 36
(c) 38 (d) 44
(e) 46

5. If a number is decreased by 4 and divided by 6, the result is 8. What would be the result if 2 is subtracted from the number and then it is divided by 5?

- (a) $9\frac{2}{3}$ (b) 10
(c) $10\frac{1}{5}$ (d) $11\frac{1}{5}$
(e) None of these

6. A number when multiplied by 13 is increased by 180. The number is (P.C.S., 2004)

- (a) 5 (b) 12
(c) 15 (d) 45

7. The sum of twice a number and three times of 42 is 238. What is the sum of thrice the number and two times of 42? (Bank P.O., 2010)

- (a) 245 (b) 250
(c) 252 (d) 264
(e) None of these

8. If one-third of one-fourth of a number is 15, then three-tenths of that number is

- (a) 35 (b) 36
(c) 45 (d) 54

9. The difference between $\frac{3}{5}$ th of $\frac{2}{3}$ rd of a number

and $\frac{2}{5}$ th of $\frac{1}{4}$ th of the same number is 288. What

is the number? (Bank P.O., 2006)

- (a) 850 (b) 895
(c) 955 (d) 960

10. A number is doubled and 9 is added. If the resultant is trebled, it becomes 75. What is that number?

- (a) 3.5 (b) 6
(c) 8 (d) None of these

11. Three-fourth of a number is 60 more than its one-third. The number is

- (a) 84 (b) 108
(c) 144 (d) None of these

12. A number whose fifth part increased by 4 is equal to its fourth part diminished by 10, is (SNAP, 2010)

- (a) 240 (b) 260
(c) 270 (d) 280